Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	27 June 2026
Project Name	Opticrop:Smart Agricultural ProductionOptimization Engine
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form Registration through Gmail Registration through LinkedIn
FR-2	User Confirmation	Confirmation via Email Confirmation via OTP
FR-3	Soil & Environmental Data Input	Manual entry of N, P, K, temperature, humidity, pH and rainfall values Auto-fetch real-time weather/rainfall data via Weather API Bulk upload of soil-test data via CSV
FR-4	Crop Recommendation Generation	Run ML model on submitted soil & climate parameters Display top recommended crop(s) with confidence score Show suitability score for a specific crop selected by the user
FR-5	Research & Analytics Dashboard	View historical crop-environment trends and patterns Filter data by region, season, or crop type Export analysis reports as PDF / CSV
FR-6	Notifications & Alerts	Email/SMS delivery of crop recommendation report Alert farmer when input conditions are unsuitable for any crop

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	Simple, intuitive dashboard that lets farmers with limited technical knowledge enter inputs and understand recommendations easily; supports regional languages.
NFR-2	**Security**	Encrypted data transmission (HTTPS/SSL), role-based access control, secure storage of user and soil/crop data, input validation against malicious data.

NFR-3	**Reliability**	Consistent and accurate crop recommendations with proper error handling for invalid or missing input values; automated model validation before deployment.
NFR-4	**Performance**	Crop recommendation results should be generated within 2-3 seconds of submitting input parameters, even under concurrent user load.
NFR-5	**Availability**	System should be available at least 99.5% of the time via cloud-based deployment with monitoring and automated failover.
NFR-6	**Scalability**	Architecture should support a growing number of farmers/researchers and expanding datasets without degradation in response time.